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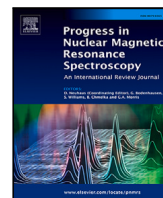


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Additive fabrication for NMR probe builders

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ABSTRACT

Three-dimensional (3D) printing has emerged as a transformative technology for nuclear magnetic resonance (NMR) instrumentation, offering flexibility in the design and fabrication of custom tools that enhance experimental capabilities. Additive manufacturing has made it possible for many NMR labs to build their own magic angle spinning assemblies, sample handling devices, and other critical components. We summarize common 3D printing techniques, such as fused deposition modeling (FDM) and stereolithography (SLA) for polymers, along with metal printing methods like selective laser melting. By facilitating rapid prototyping, 3D printing accelerates the development and optimization of NMR systems, as well as bypassing traditional manufacturing constraints. This review also discusses perspectives on the future of 3D printing in NMR and related methods, providing cost-effective, in-house solutions that increase participation, allow for sharing and remixing of innovations, and broaden applications across chemical, biological, and materials research.

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1. Introduction

The long-term prominence of NMR in chemical and biophysical applications is primarily due to its flexibility. NMR can be used to obtain atomic-level detail about the structures and motions of molecules in physical states ranging from isotropic liquids to rigid solids, as well as everything in between, where many real-world problems reside. Maintaining and expanding the applicability of NMR to ever more lifelike biological systems and practical materials applications requires continued innovation in methodology. A bottleneck currently limiting progress is the development of NMR instrumentation. NMR probes are presently highly specialized, expensive pieces of equipment with limited commercial sources and even fewer avenues for obtaining modified or customizable parts. In particular, the continued push toward ever higher fields coupled with the reduced competition in the NMR instrument market have accelerated standardization of instruments used for routine chemistry and structural biology applications. However, at the same time, many developments at the forefront of the field, including many that have ultimately become commercialized have come about because of innovations in individual labs.

For example, the gyrotron instrumentation that is now standard equipment for performing dynamic nuclear polarization (DNP) at low temperature under magic angle spinning (MAS) was mostly developed by R.G. Griffin and collaborators in stages over many years [1–3], with regular milestones in the form of enhanced signal on setup samples [4] as well as proteins of biological relevance [5]. After a long development period (and the efforts of M. Rosay, in particular), this instrumentation is now used by many applications-focused groups working on both biological macromolecules and materials science, and has been used to obtain results in systems that would otherwise be inaccessible due to low sensitivity, e.g., [6–11]. In particular, DNP has enabled protein NMR inside viable mammalian cells [12,13], structural investigation of the link between aberrant pigmentation and collagen degradation in alkaptonuria [14], and direct observation of protein–lipopolysaccharide interactions in the membranes of pathogenic bacteria [15].

Similarly, the development of Overhauser [16–18] and dissolution DNP [19–23], as well as other hyperpolarization methods including PHIP [24–27] and SABRE [28–31] has also been driven by academic labs. The utility of high-pressure NMR is also becoming better appreciated due to the work of J. Wand [32–34], whose apparatus has been commercialized by Daedalus, and A. Bax, who is using a different homebuilt apparatus to achieve rapid pressure jumps [35,36]. Such innovations are typically developed in individual labs and then become more widely used on the strength of their applications, often as a result of commercialization once their utility is established via proof-of-concept studies. However, this process is slow in part because of the expertise that is needed. Not every lab has the wherewithal to build new instrumentation after reading a paper about it, and the difficulty and expense of doing so preclude making bespoke instrumentation for potentially risky applications that are outside the scope of the initial demonstration. Therefore, even very exciting innovations in instrumentation can take a long time to disperse to the rest of the field.

In contrast, the situation with pulse sequences is quite different. Writing pulse sequences has a low entry barrier, since it can be done on any computer and tested relatively straightforwardly with standard NMR equipment. The code for new pulse sequences is generally made available for anyone to implement on their own instrument. This situation has led to a rich environment where pulse sequence innovations can be readily taken up by different groups immediately after publication. This not only leads to the widespread adoption of useful innovations, but also enables remixing, or copy/pasting building blocks and reusing them in different contexts. Having access to this kind of milieu where remixing of hardware elements becomes similarly easy and common would greatly speed up the development of new NMR methodology, and open the door to the next epoch of NMR, in which building and modifying application-specific devices is accessible

to a broader community. This will enable a greater diversity of new experiments and results than an expensive, standardized, one-size-fits-all solution. Increased access to instrument fabrication technology will also synergize well with the development of low-cost, portable spectrometers using permanent magnets [37,38].

In the context of laboratories that build their own instruments, components have traditionally been made using labor-intensive traditional machining methods, requiring a suite of machines and tools including lathes and milling machines that often require a dedicated space, extensive training, and a safety bystander to operate. The operator involvement has been reduced by the advent of computer numerical control (CNC) machines, which enable the fabrication of pieces that are extremely difficult or impossible to machine by hand. These processes are known as subtractive manufacturing, as layers of starting material are removed to produce the desired part. Although this methodology can produce pieces reproducibly and precisely, it is not always practically accessible for several reasons. Dedicated machine shops are often only available at large research universities and may not always be open for use by individual researchers. Working with the machinery requires someone with skill and expertise in this form of fabrication. Even with access to CNC machines and professional operators, understanding of the capabilities and limitations of this process must be factored into the design phase; researchers who are not familiar with the machining process are limited in their ability to design parts that can be made efficiently. In this fabrication modality, parts are made individually, so if a mistake is made or a prototype dimension is incorrect, then time must be reinvested in trying again. Although CNCs are automated, these machines are expensive and also require lengthy training for proficient operation, which is not practical for students who have limited time or labs that do not specialize in instrumentation projects.

3D printing is a form of additive manufacturing where layers are sequentially assembled to form the desired piece. Although this process is not without challenges, the technique is generally more accessible than traditional machining. One similarity is that there are different styles of printers to meet a variety of needs (and therefore no “one-size-fits-all” solution); however, a collection of several machines is unlikely to span more than half of a typical lab benchtop. Shared 3D printers are now commonplace in many engineering laboratories or other university-wide maker spaces. 3D printing software is generally free, meaning parts can be shared without the need for expensive computer-aided design (CAD) software. If it is necessary to design a new part or make modifications to an existing file, introductory versions of CAD software are also available for free, such as Tinkercad by Autodesk. The cost of many commercially available printers with print platform areas ranging from 400 cm² to 800 cm² is also comparable to benchtop mills or lathes that are capable of making parts of similar dimensions. With regard to safety, 3D printing poses lower risk to the user because printers rarely have components that could cause significant physical harm to the user and the additive process does not create sharp, hot metal chips or particulate matter. By virtue of the limited risks, the technique can be self-taught through trial and error and performed independently with no need for a safety observer. However, depending on the specific process used, 3D printing does have its own safety considerations, which will be discussed in the individual sections for each type of printer.

The layering process of additive manufacturing enables complicated pieces to be made at lower cost and with a broader scope of possible features, such as no-access internal channels or no-assembly interlocks. This has the potential to remove barriers to involvement in design at any level, and also to encourage established builders to feel comfortable investing in new, high-risk, or single-application ideas. 3D printing has also expanded past plastic filament printing; metal printing is now possible via layering techniques that mimic UV-cured resin SLA but use high-powered lasers. Examples include selective laser melting (SLM) and powder bed fusion (PBF) printing. Although the 3D printing makerspace offers a wide range of materials, only certain materials with

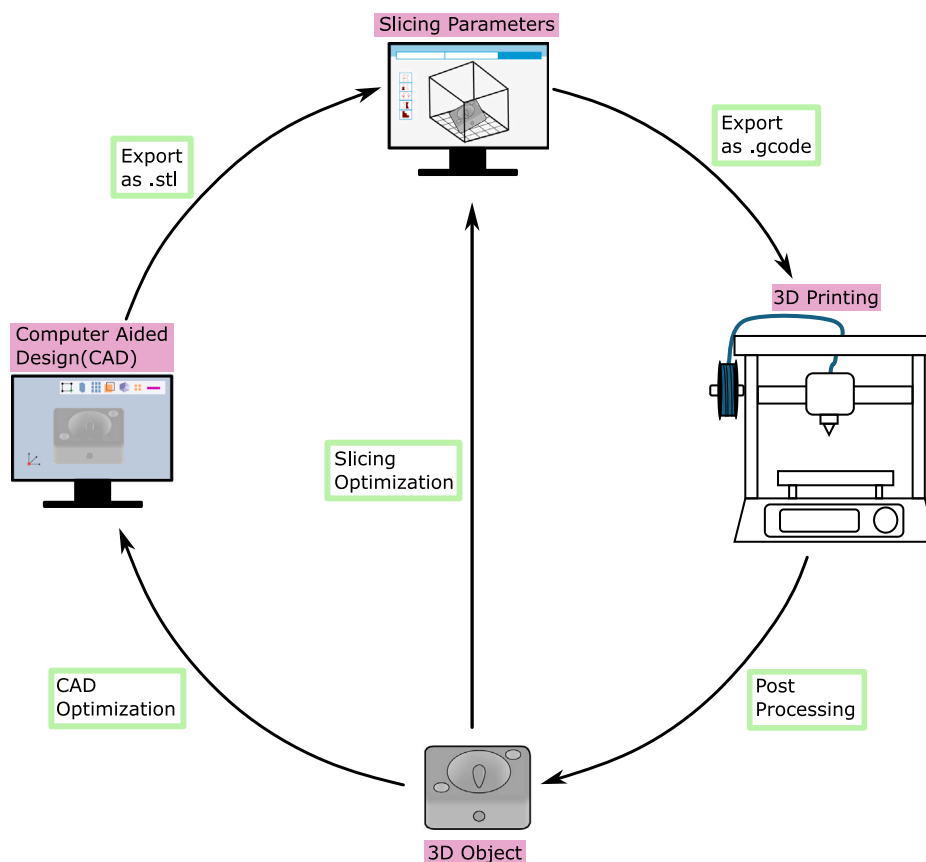


Fig. 1. Diagram illustrating the 3D printing workflow for component fabrication. The process starts with designing a CAD model, which is then sliced into printer-specific layers for 3D printing. Observations of the printed output guide iterative refinements to both the CAD design and printing parameters, optimizing performance and precision.

specific chemical, physical, and electrical properties can be selected for use in NMR instrumentation. In this review, we provide an overview of the major types of additive fabrication, discuss their respective advantages and disadvantages, and review 3D-printed instrumentation developed by NMR and EPR groups that has enabled novel designs, low-cost repairs, and shareable equipment.

2. The 3D printing workflow

Additive manufacturing (AM), commonly known as 3D printing, begins with a creative or functional design idea that is modeled in three dimensions using Computer-Aided Design (CAD) software. This software employs parametric modeling, where each element of the design is defined by spatial parameters including dimensions and angles, allowing for dynamic adjustments throughout the design process. Once the model is refined, it is converted into the STL (Standard Tessellation Language) file format, which delineates the object's surface geometry via tessellation into a mesh of triangles or facets. This creates planes parallel to the build plate, defining how the object will be built up layer by layer. Before it is sent to the printer, the STL file must be imported into a slicing software. Many slicing software options are available and they all have similar approaches to preparing the design for printing; however, most 3D printers require specific ones for compatibility. The actual printing is achieved by adding material, whether it is plastic, resin, or metal, layer by layer, building the part in stages from the ground up with respect to the build plate.

For the majority of do-it-yourself (DIY) home users, 3D printing represents a highly accessible, plug-and-play manufacturing technology, but this is dependent on making items without stringent tolerance requirements. For many hobby applications, parts are ready for use immediately after they are removed from the build plate, with minimal,

if any, post-processing required. This ease of use has significantly contributed to the recent surge in the popularity of additive manufacturing. However, for applications demanding exacting dimensional standards, more planning and post-processing may be needed, and a systematic approach to testing is required. In design for additive manufacturing, components must be designed with careful consideration of the nuances associated with the specific manufacturing process, including the material properties, effects of gravity, potential dimensional changes post-print, and post-processing requirements. Despite these difficulties, the rapid prototyping capability inherent to 3D printing provides an opportunity for continuous refinement of design features and printing parameters to ensure structural integrity and functionality (Fig. 1). For example, 3D printing was an important component of a recent MRI “hackathon”, where novice users designed and built a low-field MRI system from scratch in only four days [39]. In labs that are focused on instrumentation design, this iterative process allows for the practical enhancement of both part design and manufacturing techniques, adapting swiftly to feedback and evolving requirements. For designs that are intended for widespread distribution and use by beginners, careful troubleshooting and detailed instructions for implementation are required [40].

3. Common types of 3D printing methodology

3.1. Vat photopolymerization: SLA, MSLA, DLP

Stereolithography (SLA), masked stereolithography (MSLA), and digital light processing (DLP) are vat photopolymerization techniques where a liquid photopolymer resin is cured layer-by-layer using ultraviolet (UV) light to produce high-precision 3D-printed parts. In SLA, a laser scans a focused beam across the resin surface, selectively

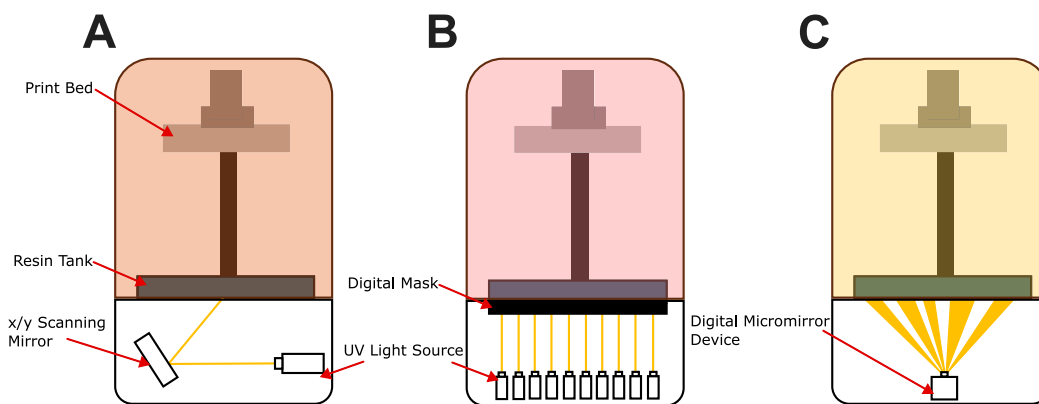


Fig. 2. Internal mechanisms of vat photopolymerization 3D printers, highlighting the focusing of UV light to form printed layers. (A) Stereolithography (SLA) uses a laser to cure resin point-by-point. (B) Masked Stereolithography (MSLA) employs an LCD mask to selectively cure entire layers. (C) Digital Light Processing (DLP) projects UV light patterns to solidify layers simultaneously.

solidifying it with a resolution down to $\sim 25\ \mu\text{m}$ in the xy plane and layer heights as fine as $10\ \mu\text{m}$. MSLA improves throughput by projecting an entire layer's pattern via a UV LCD mask, while DLP uses a digital micromirror device to flash a full-layer image, achieving similar resolution (typically $35\text{--}50\ \mu\text{m}$ xy), but with faster print times due to parallel exposure – albeit at the expense of potential pixelation from the micromirror grids. Available resins include standard photopolymers (e.g., acrylates) for prototyping, tough resins (e.g., Formlabs (Somerville, MA) Tough 1500, with tensile strengths $\sim 50\ \text{MPa}$) for durable NMR tools, and specialized biocompatible or high-temperature resins (up to $200\ ^\circ\text{C}$). Chemical compatibility with NMR samples (e.g., avoiding leaching into hydrated biomolecules or battery solutions) is critical and may require testing for individual applications, as the manufacturers' compatibility data often do not cover all of the situations encountered in a typical NMR lab. Limitations include anisotropic mechanical properties due to layer-wise curing (weaker interlayer bonding), support structure requirements for overhangs, and resin viscosity constraints. High-viscosity formulations can impede thin-layer spreading, capping practical z -layer thicknesses and slowing prints [41]. For solid-state NMR applications, such as rotor packing tools, these methods excel at sub-millimeter precision but face challenges with long-term wear under high g -forces or solvent exposure (see Figs. 2 and 3).

Handling SLA resins requires attention to chemical hazards. Uncured photopolymers, such as acrylates or epoxies, are skin and respiratory irritants that can potentially cause dermatitis or sensitization with prolonged exposure. Volatile organic compounds (VOCs) released during printing or isopropyl alcohol rinsing require adequate ventilation to reduce inhalation risks [42]. These photoreactive liquids require careful storage to avoid unintended curing, and they pose spill risks that could contaminate NMR samples or equipment in a lab environment. Post-curing with UV light generates heat, presenting a minor burn hazard, and incompletely cured parts may leach residual monomers into their surroundings, which is relevant for NMR sample-handling tools contacting biological samples [43]. These hazards can be mitigated by using personal protective equipment (PPE), gloves and safety glasses at minimum, and consulting Safety Data Sheets (SDS) for specific resins (e.g., Formlabs Tough 1500 or Clear V4) to address formulation-specific risks.

Aside from safety issues, users of stereolithography (SLA) must anticipate and mitigate several design challenges inherent to the process. Surface tension often causes liquid resin to creep into small gaps, 90° angles, or support structures, leading to excess material that hardens during curing if not removed. In order to address this, parts can be reoriented during printing to confine rounding or imperfections to non-critical areas. Similarly, concave surfaces tend to flatten post-curing due to resin flow and polymerization. This can cause unexpected results

at first, but can be readily accounted for in the design process after some experience. In the xy plane, thin features are prone to broadening as the UV laser's spot size (typically $50\text{--}100\ \mu\text{m}$) exceeds the intended footprint, necessitating oversized design adjustments. Post-processing in the form of washing, drying, and UV curing further complicate precision, as resin shrinkage (often $2\%\text{--}5\%$ depending on polymer chemistry) tightens the molecular structure, reducing dimensions and significantly impacting features like threads, where intended fits between threaded stackable components can fail unless they are compensated by preemptive scaling in the CAD model.

3.2. Materials extrusion: FFF/FDM

Fused Deposition Modeling (FDM), interchangeably termed Fused Filament Fabrication (FFF) is the most common type of 3D printing. It has been used to create a variety of custom labware, bioreactors, and other useful devices for chemistry and biology laboratories [44–47]. In FDM, parts are built by extruding molten thermoplastic filament through a heated nozzle, depositing it layer-by-layer onto a build platform. Unlike vat photopolymerization methods such as SLA, FDM relies on solid filament feedstocks, commonly polylactic acid (PLA), acrylonitrile butadiene styrene (ABS), or polyethylene terephthalate glycol (PETG). The feedstock is melted at temperatures between $180\text{--}260\ ^\circ\text{C}$, depending on the material, and solidified as the nozzle traces a digitally defined path. This process offers affordability, material versatility, and ease of use, making it a popular choice for prototyping NMR tools like autosampler components or rotor marking guides. Resolutions typically range from $100\text{--}400\ \mu\text{m}$ in layer height and xy precision, coarser than SLA but sufficient for larger, mechanically robust parts. FDM's ability to produce durable, lightweight fixtures with materials like PETG (with a glass transition temperature of $\sim 85\ ^\circ\text{C}$) supports a variety of lab workflows, including making replacement parts for small devices; however, thermal and dimensional constraints limit its use for sub-millimeter precision parts like rotor packing tools and MAS assemblies.

Safety considerations for FDM printing include high temperatures and potentially toxic material emissions, and care must be taken to avoid contamination and maintain NMR sample integrity. The heated extruder ($180\text{--}260\ ^\circ\text{C}$) and bed ($50\text{--}110\ ^\circ\text{C}$) pose burn risks, requiring users to avoid direct contact during operation and cooldown, while filament off-gassing – especially from ABS – releases volatile organic compounds (VOCs) such as styrene, necessitating a well-ventilated space or an enclosure with HEPA filtration [48,49]. Fluorinated filaments, such as polyvinylidene fluoride (PVDF) or polytetrafluoroethylene (PTFE-like) blends, present additional hazards; when overheated beyond their recommended range ($240\text{--}300\ ^\circ\text{C}$), they can decompose, emitting toxic

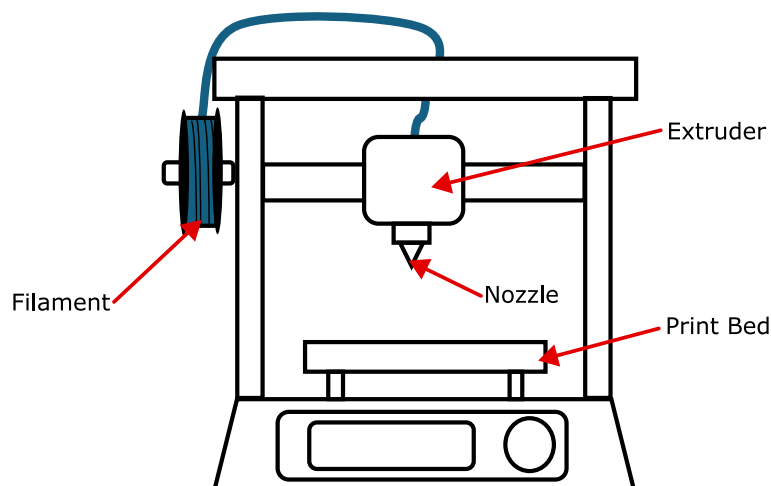


Fig. 3. Illustration of a generic Fused Deposition Modeling (FDM) 3D printer, showcasing key components. The nozzle deposits molten filament, driven by the extruding motor, which feeds the filament from a spool onto the print bed, where the object is built layer by layer.

gases including hydrogen fluoride (HF), a severe respiratory and skin irritant [50,51]. In our own laboratory, the printer is placed in a fume hood when working with fluorinated filaments. For NMR labs, food-grade or low-emission filaments (e.g., PLA) should be used for tools that will come in contact with sensitive samples, alongside consulting Safety Data Sheets (SDS) for specific materials – e.g., Ultimaker ABS (Utrecht, Netherlands) or Nile Polymers Fluorinar-C PVDF (Centerville, UT) – to verify extrusion temperatures and chemical compatibility. In addition to adequate ventilation, basic personal protective equipment (PPE), including heat-resistant gloves and safety glasses, further ensures safe handling during printing and post-processing.

FDM's ease of use is tempered by printing limitations that users must navigate, especially for applications demanding precision or durability. Layer adhesion and anisotropy arising from the sequential deposition of molten filament yield parts with reduced strength along the z -axis (up to 40% weaker than in the xy plane), which is a drawback for components under mechanical stress. Overhangs exceeding 45° or thin features below ~ 0.8 mm in the xy plane often require support structures, which may leave surface imperfections or mar delicate geometries unless meticulously removed, while bridging gaps wider than ~ 10 – 20 mm risks sagging due to filament cooling dynamics. Thermal shrinkage (0.5%–2% for ABS) and warping, both of which are worsened by uneven cooling in larger prints, can misalign critical features, necessitating design adjustments, such as scaling models or adding chamfers to ensure proper fits. A range of nozzle diameters, from 0.1 mm to 1.8 mm, is available; finer nozzles (e.g., 0.1–0.4 mm) enhance detail for intricate designs, though at the cost of slower print times, whereas larger ones (e.g., 0.8–1.8 mm) prioritize speed and strength for bulkier parts. Post-processing, such as sanding or acetone smoothing for ABS, can further improve surface quality and dimensional accuracy across these applications, albeit at the expense of additional time spent on each part and a potential loss of reproducibility.

3.3. Metal and lost wax casting by SLA

Metal 3D printing comprises a suite of additive manufacturing techniques that enable the creation of intricate metallic components [52]. Selective Laser Melting (SLM) and Selective Laser Sintering (SLS) are powder bed fusion processes: SLM employs a high-powered laser (typically 200–400 W) to fully melt metal powders such as titanium, stainless steel, or aluminum alloys into dense [53], robust structures ($>99\%$ density) with mechanical properties rivaling conventionally

forged parts. Methods are being developed to improve resolution by reducing the spot melting size, which is difficult with copper's high conductivity [54]. SLS partially sinters powders, producing porous, lightweight forms that are suitable for specialized uses like thermal management. Lost wax casting, which is a traditional jewelry making technique, has also been adapted for 3D printing. Here, FDM or SLA is used to craft a wax or polymer pattern that is encased in a ceramic mold, burned out, and replaced with molten metal (e.g., bronze, gold, or steel), merging the design freedom of additive manufacturing with the precision of classic casting. Because they can produce resolutions as fine as 20–50 μm (SLM and SLS) and superior surface finishes (lost wax casting), these methods excel at producing complex geometries such as lattices or internal channels that are unattainable through machining. This aspect of 3D printing methodology in particular has the potential to speed up the process of teaching new users to build NMR probes, because much of the training required when designing for traditional machining focuses on designing parts that can be fabricated by a step-by-step subtractive process.

In contrast to the powder bed fusion and casting methods described above, the Virtual Foundry (Stoughton, WI) offers an innovative approach that combines some of the advantages of FDM and sintering. Its copper-infused filament can be extruded using standard FDM printers and then sintered in a furnace to produce a fully metallic copper part. This hybrid process uses the widely accessible FDM technology to produce the basic shape followed by a sintering step that eliminates the binding material and fuses the copper particles, making metal 3D printing viable for users without access to specialized equipment. Similarly, Markforged's (Waltham, MA) Metal X system utilizes FDM to print with its Copper 200cc spool, a copper bead-infused filament (other materials such as stainless steel and titanium are also available), followed by a process that washes away the binder with a proprietary solution. The part is then baked to form a solid copper structure that can achieve up to 84% IACS (International Annealed Copper Standard) electrical conductivity [55]. Like the Virtual Foundry's method, this approach incurs slight shrinkage from binder removal during post-processing, necessitating design compensation to maintain dimensional precision. These techniques are highlighted here because many metal components in NMR probes, demand high-conductivity materials for optimal radiofrequency (RF) performance. These FDM-based methods offer a potential alternative to traditional machining by enabling complex, conductive copper parts with reduced manufacturing constraints [56].

4. Solid-state NMR probe heads

4.1. Static NMR

Many useful experiments can be performed on static samples in both solid-state and solution NMR. Static NMR is valuable for samples such as large single crystals [57,58], oriented membranes [59,60], and live animals [61,62]. Single-crystal static NMR allows researchers to gather orientation-dependent structural information by directly probing the crystal lattice in defined orientations relative to the external magnetic field (B_0). Purpose-built single-crystal NMR probes commonly utilize a goniometer in the probe head to rotate a crystal to defined angular positions relative to B_0 with high precision for better data correlation in rotation plots [63].

A three-axis goniometer probe was constructed with a dovetail tenon for crystal mounting; it is compatible with a modified X-ray diffraction goniometer, making it possible to obtain high-accuracy correlative data for hybrid NMR-crystallography data analysis [64]. The probe design was later altered to a two-axis goniometer that was placed outside of the RF coil, improving the filling factor and RF sensitivity [65]. This version used a compromise axis of rotation, at 45° to B_0 rather than 90°, for a more compact probe head configuration. With access to 3D printing, Vinding et al. created a modular single-crystal goniometer kit that could mount to an existing wide-bore NMR probe with minimal alteration [66]. The goniometer was modularized to allow modifications for use with different probes, and the parts were designed to offer reproducible and durable prints. In the original study, an 8 mm diameter goniometer was placed inside of an 11 mm inner-diameter ribbon solenoid. The plastic components were manufactured using an Ultimaker 2 FDM printer using PLA. Commercial goniometer probes exist and may be the optimal solution for applications where the crystal sizes are relatively uniform and the range of desired orientations is the same for each experiment. However, 3D printed versions offer greater flexibility to tailor the sample housing (and potentially the RF coil) to match the crystal size for the particular samples under investigation, and to choose the range of accessible orientations based on the specific NMR experiments planned.

4.2. Magic angle spinning modules

In solid-state NMR, interactions that have an anisotropic component with respect to the B_0 field (e.g., chemical shift anisotropy and homonuclear or heteronuclear dipolar couplings) are typically scaled by spinning at the magic angle. MAS simplifies the spectra of solids considerably, but introduces some particular instrumental requirements. In general, a probe head designed for MAS incorporates a spinning module or rotor housing as its primary component. As 3D printing of spinning modules has become more commonplace, there has been increased attention to measuring the magnetic properties of filaments and resin materials. In the imaging community, the focus is on generating 3D-printed phantoms that mimic particular properties of biological tissue, such as density or local discontinuities in magnetic susceptibility [67,68]. Sangal et al. have systematically evaluated the magnetic susceptibilities of commonly used 3D printing filament and resin materials [69]. Using a 3T MRI scanner, they mapped local magnetic field variations induced by these materials in the B_0 field, combining experimental measurements with theoretical calculations, using exact analytical solutions for cylindrical geometries, to precisely quantify susceptibilities, validated against the known susceptibility of air. Their findings are compiled in MaDaMEPro (Material Database for Magnetic Properties) [69], an open-access, comprehensive resource that not only details magnetic susceptibilities but also includes 3D printing parameters and printer specifics. Schönzart et al. have recently developed MAS modules that are optimized for minimizing perturbations to the B_0 field, by enhancing the symmetry of components to the extent possible and by minimizing sharp changes in magnetic susceptibility close to the

sample. In the course of this work, they have measured the magnetic susceptibilities of many materials used in the construction of NMR probes, including common 3D printer resins [70].

In modern NMR probes, spinning is typically achieved pneumatically: air bearings suspend the rotor while a drive stream propels it to high rotational speeds. The application of the drive mechanism varies depending on the rotor type; for pencil rotors, it is applied at the bottom end of the stator (Fig. 4C), whereas for flat-bottom rotors, it is applied at the top end (Fig. 4B). The spinning module comprises several finely machined components (Fig. 4A), crafted to extremely tight tolerances, sometimes within the single-digit micron (μm) range [71]. These precise dimensions ensure an air-tight seal upon assembly of the spinning module. The spinning module's design features dedicated airways that direct high-pressure air either to the bearings in a double-bearing configuration or directly to the drive, thereby enabling fast, stable rotation. The rotors spinning within these spinning module must be manufactured with equally precise tolerances. Typical materials are yttria-stabilized zirconia, silicon nitride (Si_3N_4), and sapphire. Although limited information is available from manufacturers regarding commercial rotors, many published designs are available. A recent study demonstrated the fabrication of 0.7 mm (700 μm) diamond rotors with tolerances of $\pm 5 \mu\text{m}$ [72], which are comparable to the tolerances measured for commercially available 0.7 mm zirconia rotors. Regardless of the rotor material, critical dimensions including tolerances, concentricity, cylindricity, straightness, and surface finish must be *near perfect* to ensure optimal spinning performance and prevent damage. This precision becomes increasingly vital as rotor miniaturization advances and higher MAS rates exceeding 100 kHz are pursued.

In general, rotor crashes are the most commonly encountered hardware problems in solid-state NMR. If a rotor crashes and shatters at high MAS rates, irreparable damage is often caused to the spinning module itself. The RF coil, due to its proximity to the spinning rotor, can suffer from mechanical deformation or electrical short-circuiting, necessitating its replacement as well. The suspension, or radial, bearings, that maintain the rotor's position, are also highly susceptible to damage from rotor fragments, leading to compromised integrity or complete failure. The destruction of these components not only halts the NMR experiment but can also contaminate the probe with debris, potentially affecting subsequent experiments even after repair. Fabricating these parts in-house can significantly reduce downtime associated with purchasing commercially, thereby minimizing overall costs and speeding up the return to operational status. In-house fabrication requires careful attention to materials science, mechanical engineering, and NMR physics to prevent introducing new failure points or performance inconsistencies compared to commercial products.

Repairing a probe after a rotor crash requires:

- Precision fabrication of components to match or exceed commercial standards.
- Use of materials that can withstand the mechanical stress and chemical environment within the probe.
- Calibration of air flow systems to maintain optimal bearing performance and rotor stability.
- Quality control to verify that each spinning module component adheres to the stringent specifications required for high-speed MAS.

Despite the challenges, in-house fabrication of MAS spinning modules and associated components is one of the most common applications of 3D printing in NMR. These are summarized in Table 1. For example, Banks et al. [71] produced 3D printed MAS spinning modules, replacing many components with 3D printed parts while maintaining performance. The main body of the module was 3D printed from ZrO_2 using a 3DCeram Sinto (Limoges, France) printer. Plastic components, including gas inlet bushings, coil block, bottom plate, and fiber optic

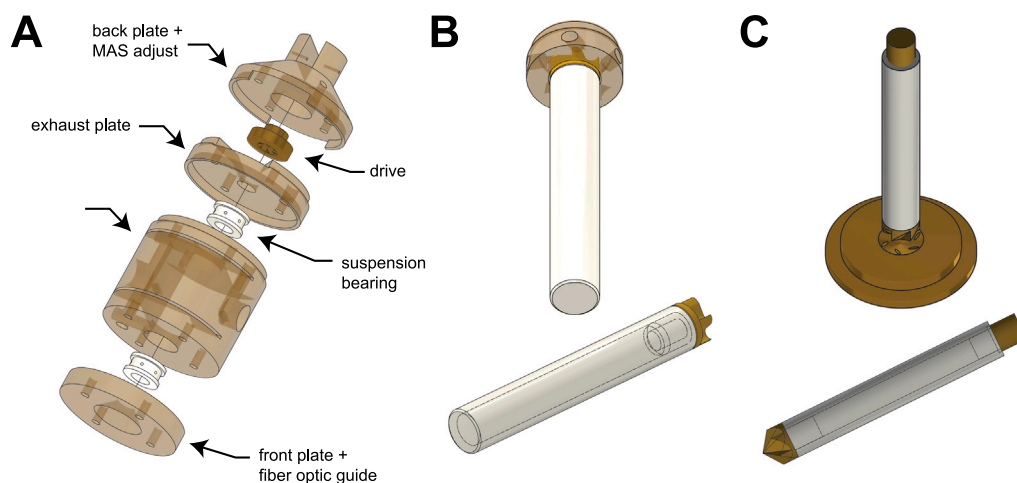


Fig. 4. (A) Exploded view of MAS stator assembly for pencil rotors, with the critical components labeled. (B) Isolated view of the flat-bottom rotor inside the drive plate and (C) a pencil rotor above the conical drive mechanism.

Table 1

Comparison of the 3D printing methods and materials utilized in the fabrication of various MAS NMR spinning assemblies, along with the rotor types supported by each spinning module and their experimentally determined maximum stable drive speeds [71,73–75].

Print method	Material(s)	Accommodated rotor type	Rotor diameter (mm)	MAS rate	Reference
FDM	ABS	pencil	7	5 kHz	[73]
			4	10 kHz	
			3.5	23 kHz	
SLA	Clear V4 Microfine Green ZrO ₂	Flat-bottom	3.2	>30 kHz	[71]
SLA	Craftsman	Flat-bottom	4.0	12 kHz	[74]
FDM	ABS	conical	14	3.9 kHz	[75]
			3.9	20 kHz	

block, were printed with a Formlabs Form 3 SLA using clear resin. The drive plate used Protolabs' SLA (Maple Plain, MN) MicroFine green resin for its tighter tolerances. However, due to the precision required for radial bearings, these were machined rather than printed, given the limitations of ceramic 3D printing (± 0.1 mm tolerance, 15%–30% shrinkage, and lower resolution). The printed stators achieved MAS rates over 30 kHz, with costs around \$2,500 per spinning module, primarily due to the machined ceramic parts.

Pereira et al. [74] further reduced costs by designing a “one-shot” 4 mm stator, inspired by both the Bruker and Banks et al. designs [71]. They integrated the spinning module body, bearings, drive plate, axial bearing, and Bernoulli valve into a single part printed with an Anycubic Photon M3 Plus SLA printer (Shenzhen, China) using their Craftsman resin. This approach eliminated potential air leaks and alignment issues. The printer offers a z-axis resolution of 10 μ m, allowing for varying fine air holes, 0.5 mm being the smallest ones. The stator reached a maximum operational spin speed of 12 kHz, requiring higher drive pressure than its traditionally machined counterpart due to printing-related imperfections in the drive jet holes. This design significantly lowered the production cost to about \$5 per stator, which is less than 1% of commercial costs.

Despite FDM 3D printing's limitations in achieving micron-level tolerances, Xu et al. pioneered a fully 3D printed spinning module using ABS filament [73]. Their design included an all-in-one housing with a double-bearing configuration and a drive nozzle tailored for pencil rotors. The printing process for each module took 5–8 h, varying

with rotor size. To accommodate FDM's constraints, they used teardrop shapes for bearing holes to prevent overhang and achieve circularity. Stators were produced for 7 mm, 4 mm, and 3.5 mm rotors, achieving stable spin speeds of 5 kHz, 10 kHz, and 23 kHz, respectively.

High-resolution benchtop NMR typically focuses on liquid samples due to the simplicity of the probe head design, which often consists of a small cavity containing a stationary one-turn transverse saddle coil. The spinning mechanisms for MAS are usually too large for permanent magnet assemblies, limiting most benchtop solid-state NMR to static powder sampling. However, Xu et al. [75] used 3D printing to scale down and modify a spinning module design to enable MAS in space-constrained settings. They developed a miniaturized conical rotor and module assembly, initially tested at a macro scale with 83 modifications and 7 redesigns, highlighting the utility of rapid prototyping. After optimizing the design, they scaled down to a 9.5 mm spinning module and a 3.9 mm outer-diameter (o.d.) conical rotor. The stator was FDM 3D printed using ABS, and the rotor with polycarbonate (PC) filament, with a second stator printed in zirconia ceramic for improved quality. This material choice resulted in a 28% shrinkage during debinding and annealing, yielding a preferable stator geometry. This system allows spinning of a 2.65 μ L NMR sample to speeds over 20 kHz.

4.3. Rotor drive caps

Rotor drive caps play a crucial role in achieving the high rotational speeds required for MAS. Table 2 presents a summary of the 3D printed drive caps discussed in this section. These caps, often referred to as turbine caps, convert pneumatic pressure into mechanical rotation by directing airflow through a series of carefully designed fins. The geometry and orientation of these fins are critical; they must be optimized to maximize torque while minimizing air turbulence and energy loss. Key design considerations include the angle, length, and number of fins, all of which directly influence the efficiency and stability of rotor spinning. Proper fin design ensures that the rotor can reach and maintain the high speeds necessary for effective MAS, where even slight imperfections can lead to decreased performance or rotor instability. Drive caps are engineered to fit snugly into the cylindrical rotor, ensuring a secure connection that prevents air leakage and maintains rotational symmetry. This precise fit is essential for the rotor's balance and the integrity of the spinning process. Commercially available rotor drive caps come in various materials, with Torlon polyamide-imide and Vespel polyimide among the most commonly used. Others include zirconia for high-performance applications and polychlorotrifluoroethylene (PCTFE), also known as Kel-F, for its chemical resistance and lack of proton background.

Table 2

Comparison of the material, size, and maximum MAS rates of 3D-printed NMR rotor drive caps from recent studies [71,73,76].

Print method	Material	Rotor type	Diameter (mm)	MAS rate	Reference
FDM	polycarbonate	Pencil	3.5	18 kHz	[73]
FDM	polycarbonate	Pencil	4.0	10 kHz	[73]
SLA	Microfine Green	Flat-bottom	3.2	30 kHz	[71]
SLA	Clear V4	Flat-bottom	1.3	46 kHz	[76]
SLA	Clear V4	Flat-bottom	2.5	26 kHz	[76]
SLA	Clear V4	Flat-bottom	3.2	22 kHz	[76]
SLA	Clear V4	Flat-bottom	4.0	10 kHz	[76]
SLA	Clear V4	Flat-bottom	7.0	6 kHz	[76]

Xu et al. [73] demonstrated the use of 3D printing for drive caps, crafting 4 mm and 3.5 mm caps from PC, which enabled MAS rates of 10 kHz and 18 kHz, respectively, matching the performance of traditionally machined components. They also developed a single crystal rotor insert, for 4 mm rotors, printed using ABS, that has an integrated contrast mark for optical sensors. This insert, when used with a Revolution NMR 4 mm pencil rotor, achieved a spinning speed of 10 kHz.

In another study, Banks et al. had 3.2 mm drive caps manufactured by Protolabs using MicroFine green resin, an ABS-like material [71]. These caps, costing \$20 each, reached MAS rates over 30 kHz, closely matching commercial performance. Amerein et al. utilized a Formlabs Form 3+ printer with Clear V4 resin to produce drive caps for rotors from 1.3 mm to 7 mm. Larger caps (3.2 mm, 4 mm, and 7 mm) had a near-perfect replication rate and performed comparably to machined caps. However, smaller caps (1.3 mm and 2.5 mm) showed a slight performance drop due to the printer's limitations, with maximum MAS rates of 46 ± 1 kHz and 26 ± 2 kHz, respectively, compared to 67 kHz and 35 kHz for commercial models [76].

4.4. Spherical rotors

One of the earliest 3D-printed MAS module designs used spherical rotors with machined grooves on the outside to direct the flow of the drive air [77]. Spherical rotors have some advantages over cylindrical ones: some unique characteristics include (1) reduced wobbling compared to traditional cylindrical rotors, (2) a single gas stream can be used to suspend and drive the sphere, and (3) sudden rotor instability does not lead to crashing and damage to the spinning module. Disadvantages include increased difficulty of loading and unloading samples, reduced filling factor when using traditional coil designs, and lack of interoperability with commercial probes. The module that drives this rotor was printed using Formlabs' Form 2 SLA printer using Clear V4 resin. Fast prototyping was essential to the development of this stator design, with over 230 designs tested before the initial publication. Possible turbine grooves on the rotor were tested to showcase the effects of turbine geometry on spin speeds and stability [78]. A benefit of this rotor and stator design combination is that it self-adjusts to the magic angle (54.7°) due to the geometry of the stator. Sapphire (Al_2O_3) spheres (Sandoz Fils SA) with a $9.5 \pm 25 \mu\text{m}$ were purchased and machined to hollow out the sample chamber to $10 \mu\text{m}$ tolerance [79]. Two 9.5 mm o.d. spherical rotors were machined using zirconia and hollowed to different sample volumes; one matching the $36 \mu\text{L}$ sample volume of a 3.2 mm cylindrical rotor and the other to $161 \mu\text{L}$ to improve filling factor and signal-to-noise (SNR). The rotor reached stable spinning frequencies of 4.6 kHz and 10.6 kHz using nitrogen and helium gas, respectively.

The performance of the probe used with this spherical rotor MAS design was improved by replacing the original split solenoid with a double-saddle coil and scaling the rotor down to 4 mm o.d. [80]. The double-saddle coil is a transverse resonator that generates B_1 RF fields perpendicular to the sample axis [81]. The hemispherical cup of the stator was thinned out to further improve the filling factor, resulting in

better signal intensity. Cross polarization (CP) experiments using this smaller 4 mm rotor were performed at 11.4 kHz; however, spinning frequencies over 28 kHz were reached using helium gas in spin tests. Thereafter, the spherical rotor was further miniaturized to a 2 mm o.d., containing $0.3 \mu\text{L}$ of sample volume [82], and the 3D printed stator redesigned to accommodate the small rotor size. Specifically, the air supply aperture opening was decreased to 0.5 mm to maintain a similar suspension to drive air ratio to the previous larger designs. These 2 mm o.d. rotors achieved greater than 68 kHz stable MAS rates when supplied with helium gas in benchtop testing. The smaller spinning assembly also allowed a smaller triple-saddle coil to be used, improving sensitivity and filling factor.

5. Excitation: transceiver coils

Coils are a key component in MR instruments, and development of new coils is an active area of research for both NMR and MRI instrumentation. A transceiver coil has the dual function of nuclear spin excitation and signal detection. The most common coil geometries for liquid- and solid-state NMR are saddle coils and solenoids, respectively, however other geometries have been developed for specific applications. Some such geometries include variable pitch solenoids [83], slotted tube resonators [84], Alderman-Grant coils [85,86], ribbon coils [87], Z coils [88], double saddle coils [81,89], and tilted solenoids [90]. Each coil geometry has its own benefits, such as reduction of sample heating for lossy biological samples [85,88], increased RF homogeneity [83,87], and increased sensitivity at the magic angle [90], just to name a few. However, as with most instrument designs, reliably fabricating almost all of the aforementioned coil geometries is challenging. They are often wound by hand using wire, or assembled using purpose-built jigs on an individual basis. Various 3D printing techniques ranging from using 3D-printed dissolvable scaffolding to printing with metal have been implemented as alternative fabrication methods.

The cost-effectiveness of FDM printing has led to the design of coil scaffolding for both NMR and MRI coils. One method used a sample tube with external ridges as the scaffolding for two different coil geometries: a variable pitch solenoid and a modified Alderman-Grant coil [91]. These outside ridges, dubbed fins, allowed for control over the placement of flat copper wire for the sample coil placement in relation to the sample holder (Fig. 5A). The increased design flexibility of 3D printing allowed customization of sample holder and coil, with different scaffolding designs for solid and aqueous bio polymers with different sample volumes. A 0.1 mm nozzle on the FDM printer allowed for a thin wall, with only ~ 0.3 mm of separation between the coil and sample, maximizing the filling factor. Both the 1D spectrum of an aligned protein sample and 2D NMR spectra of aligned bicelle samples were equivalent to or better than similar spectra obtained using traditionally fabricated coils [91]. Similarly, FDM printing was used to help fabricate a low-cost, under €750 (EUR), birdcage resonator, to maximize the inner diameter for large samples for MRI experiments [92]. PLA was used as structural support for all three components, allowing copper strips to be easily placed and the resonator to be built for minimal cost. Characterization of the BC resonator showed the coil had a relatively homogeneous B_1 field for samples with a diameter and length of 60 mm and 40 mm respectively. Further comparison of the home-built BC resonator and a commercially available resonator from Bruker showed equivalent performance based on signal-to-noise ratio and image quality [92]. Both the NMR and MRI coils were fabricated easily, reducing the cost while maintaining similar quality of spectra. Another useful feature of FDM printing is that one of its most common filaments, ABS, can be dissolved with acetone, enabling printing of sacrificial templates that can be removed without damaging the coil. Constant- and variable-pitch solenoids have been produced using a 3D-printed ABS template that is dissolved after winding [93] (Fig. 5B). Similar to other 3D printed coil templates, the scaffold can be easily

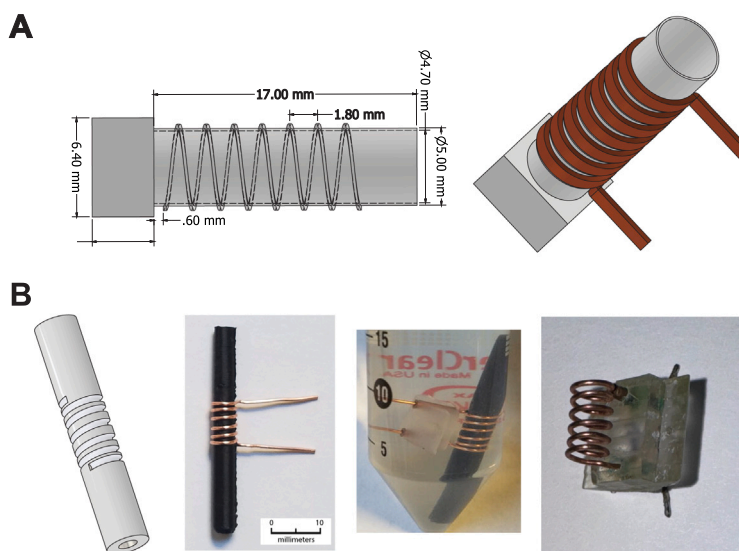


Fig. 5. (A) 5 mm o.d. finned sample chamber (left) mounted on a dual resonance solid-state NMR probe head (right) [91]. (B) Illustrates the 3D printing process for creating dissolvable ABS coil templates for NMR solenoids [93]. From left to right: CAD model design, 3D printing the template and winding the solenoid, dissolving the template in acetone, and soldering the RF solenoid onto a stator belly.

modified with little cost and effort in order to adapt the design for individual probes and sample geometries.

3D printing can also be used to produce molds for liquid metal injection for coil fabrication. Liquid metal injection molds were made using SLA and FDM for a multitude of different probehead designs, including different coil geometries such as a saddle coil and an Alderman-Grant coil [94]. One design was a continuous flow separation (CFS) probehead with a saddle coil. To determine the correct FDM filament to use, the electromagnetic properties of each material were characterized by their dielectric constant, loss tangent, and Q factor, finding that PLA had properties similar to PTFE, a commonly used non-3D printable polymer. The mold was 3D printed using FDM with PLA filament and SLA with UV curable resin, then injected with liquid gallium alloy, EGaln, to make the CFS probehead and coil. Comparison of the SNR and Q factor of the in-house probe to commercially available instrumentation showed a significant decrease in Q factor; however, an increase in SNR was observed [94]. The decrease in Q ($Q_{LM} = 44$), compared to a copper coil of the same geometry ($Q_{Cu} = 175$), was linked to the worse electromagnetic properties of gallium compared to copper, which is much more conductive. The increase in SNR was due to the improved filling factor of the custom coil geometries, illustrating how purpose-built coils can improve performance even when the electrical properties of the material are not ideal.

Expanding metal 3D printing to accessible, cost-effective methods is an active area of research in additive manufacturing. Currently, powder bed fusion is the most common method of metal 3D printing. So far, however, its printer and material cost are similar to those of traditional manufacturing; therefore, it does not have the same rapid prototyping advantages as other 3D printing methods. As mentioned in a previous section (3.3), other metal printing methods using FDM and SLA are emerging as potentially cost-effective options; however, they have yet to become a staple for metal 3D printing. Another approach to metal 3D printing is to build a custom 3D printer that combines different techniques to achieve metal prints. This methodology has been used to 3D print both NMR transceiver coils and MRI gradient coils. Constant pitch NMR transceiver coils were fabricated by using commercially available conductive ink with silver nanoparticles for screen printing fabrication and extrusion 3D printing [95].

Screen printing fabrication works by dispersing ink across a flat mesh screen that selects for the desired geometry. The ink binds to a substrate behind the mesh screen and is then sintered, leaving behind

the desired metal object. This technique has been used in manufacturing many electrochemical devices, as inks with metal nanoparticles are widely accessible materials [96]; however, this process is limited to the xy plane and cannot be used to fabricate 3D objects. To obtain the 3D shape, a 3D printer for material extrusion was modified for printing the viscous metal nanoparticle ink onto a rotating borosilicate glass tube, which acted as the substrate for the ink to adhere to in order to produce a solenoid in the z direction. A proton NMR spectrum of ethanol was collected using these coils: the J-couplings of the molecule could be discerned even at the low frequency, indicating good performance [95]. For MRI coil fabrication, a modified 3D printer combined different 3D printing extrusion based methods such as direct write, FDM, SLS, and spray coating to 3D print a silver gradient coil [97]. Software was developed to convert magnetic field specifications into G-code. To print the wire, direct-write was used to place a silver paste on an aluminum nitride substrate. Because this type of gradient coil does not require layers in the z -direction, no supports needed to be generated, unlike the case of more complex coil geometries such as saddle coils. Unlike the other 3D printing methods described here, these modified printers are not commercially available limiting the dissemination of these techniques to other labs, however they show the potential of new 3D printing methods for NMR instrumentation.

After transceiver coils are made for NMR probes, it is necessary to assess their RF homogeneity, a critical property ensuring uniform RF magnetic field distribution and consistent nuclear spin excitation across the sample. This evaluation is vital for optimizing probe performance and sensitivity. To streamline this process, the Auto-Ball Shift device, controlled by an Arduino microcontroller (Monza, Italy) and 3D-printed mechanical parts, automates the ball shift assay to map transceiver coil homogeneity directly on the benchtop [98]. This apparatus takes advantage of FDM's infill settings, which generate internal structures with adjustable patterns and densities, to produce components that are both robust and lightweight. For instance, probe mounting adapter and stepper motor holder – the largest components – were 3D printed with PLA. On the other hand, the sliding rods, requiring a smooth, durable finish, were fabricated via SLA with Clear V4 resin. This hands-free assay, combined with Python-based data processing, enhances reproducibility, minimizes human error, and speeds up coil testing, providing NMR labs with a practical solution to refine RF coil designs without heavy reliance on magnet time.

6. Sample Chambers: Inside-the-print NMR experimentation

Fused Deposition Modeling (FDM), a *dry* 3D printing technique that extrudes solid thermoplastic filament, eliminates the need for the liquid resins or powders used in stereolithography (SLA) or selective laser sintering (SLS), enabling the direct production and sealing of internal cavities and hollow segments without additional post-processing steps. In SLA or SLS, hollow chambers within printed components often retain residual uncured resin or unbound powder, requiring post-processing, such as the incorporation of drain holes, to remove these materials. In contrast, FDM produces clean, hollow structures in a single process due to the layer-by-layer deposition of a molten filament that solidifies upon cooling. This capability not only simplifies production but also helps to minimize component weight. It is also useful in cases where the internal void is designed for functional purposes, such as housing samples for spectroscopic analysis. In the context of NMR spectroscopy, this technology enables researchers to build sample containers that accommodate a wide range of sample types from volatile liquids to powders, while ensuring gas-tight seals, reducing material-related background signals, and enhancing compatibility with RF coil configurations, ultimately improving the sensitivity and resolution of NMR measurements.

FDM 3D printing has been used to make gas-tight reaction vessels, including flasks, cuvettes, and tubes of varying geometries. These were fabricated for compatibility with multiple spectroscopy techniques [99]. ABS was chosen for its mechanical stability and ease of printing, minimizing sample contamination risks, with an acrylate-balanced variant (ABS-K) used for cuvettes to reduce opacity. Printing occurred inside a glovebox under an inert N_2 atmosphere using a commercial FDM printer, and air-sensitive compounds were transferred using Schlenk lines under high vacuum. Enhanced layer adhesion improved impermeability, enabling NMR tube replicas and sealed tubes to track reactions like cyclohexanone enolization. Elongation at break tests revealed a two-fold improvement in layer adhesion under inert conditions ($\epsilon_B(N_2) = 10.7 \pm 2.6\%$ versus $\epsilon_B(\text{air}) = 4.4 \pm 1.5\%$), enhancing vessel impermeability [99]. This was further validated through limit-testing with solvents like methylcyclohexane and cyclohexane at elevated temperatures. For solution NMR, two tube designs were printed: (1) a 5 mm NMR tube replica and (2) an inert-gas-sealed tube with a spinner (Fig. 6A). ABS NMR peaks were characterized by dissolving the material in $CDCl_3$. A well-resolved spectrum of CD_3OD was taken using the 3D printed replica tube. The sealed tube, paused twice during printing to insert reagents, tracked cyclohexanone enolization via ^{13}C and ^{29}Si NMR, with spectra matching a control sample in a glass tube. Subsequent iterations adopted polyamide (PA) for enhanced solvent compatibility and minimal NMR background, and were used to track aryl naphthylalkyne synthesis [100]. Printing under an inert atmosphere allowed slightly higher nozzle temperatures without oxidation or filament degradation, ensuring print reliability. These 3 mm wall-thickness tubes, printed in approximately 3.5 h (including reagent pauses), maintained gas-tight seals under 10 bar pressure testing.

The ability to 3D print components with thin walls and tailored material properties increases the usable sample volume within a constrained space and optimizes the filling factor. These components can also incorporate multifunctional features, such as a sample chamber for powder and liquid NMR that includes an outer coil former with fins to support solenoid wrapping at a specified diameter and pitch (see section above) [91]. Two versions were designed: a simpler one for a single-wound coil and a dual-coil variant featuring an inner solenoid and an outer modified Alderman-Grant coil, a transverse coil with low-electric-field (low-E) properties designed for electrically lossy biological samples. Both designs enhance RF coil proximity to the sample, boosting RF sensitivity and signal quality. Printed using a LulzBot Mini2 (Fargo, ND) with a 0.1 mm copper nozzle, these tubular chambers achieved 0.15 mm wall thickness, yielding a sample volume of 200 μL compared to 150 μL in conventional glass tubes. A

thin silicone glue layer sealed the inner walls, preventing water and protein absorption. Adjustable chamber depth and volume, without altering coil position, allowed precise sample placement within the homogeneous RF B_1 field. After loading, the tube's inner diameter was fitted with a rubber cap, ensuring an airtight seal. Although several materials with varying performances were tested, PETG was preferred for NMR [91]. Organic polymer filaments, rich in natural abundance 1H and ^{13}C , can introduce background signals in ^{13}C NMR. Likewise, Nylon, an aliphatic polyamide with protein-like N–H bonds resembling protein backbones [102], may complicate experiments involving related nuclei, though it offers superior print quality and ease of printing. An ^{15}N -labeled proline sample, analyzed via 1D and 2D 1H -decoupled ^{15}N NMR, showed higher sensitivity and SNR than traditional methods [91]. To demonstrate adaptability, a rectangular PETG chamber for single-crystal samples optimized the filling factor, enabling a smaller “flat” coil with enhanced RF efficiency and superior 2D NMR resolution compared to a traditional apparatus with a larger coil.

A dissolvable mold was also used to make a microfluidic device of a connected sample container and transceiver coil for NMR, named Embedded SCAffold RemovinG Open Technology (ESCARGOT) [101]. Polydimethylsiloxane (PDMS) is a common material used to create the mold for microfluidic devices as its solidification can be controlled through ultraviolet light. However, a limitation of PDMS is that harsh conditions are needed for the removal of the scaffolding for the fluid channel. By making the scaffolding out of ABS, the fluid channel can be easily dissolved in acetone allowing for external components, such as an NMR transceiver coil, be added before the PDMS is cured. Round copper wire was wrapped around the ABS scaffolding and placed into uncured PDMS. Once the PDMS was cured and ABS removed, an interconnected sample chamber and solenoid transceiver coil could be used for small volume NMR [101].

Researchers have advanced NMR spectroscopy at sub-nanoliter (sub-nL) scales by employing a high-resolution 3D printing technique – two-photon polymerization – to fabricate microfluidic structures that enable precise handling and analysis of samples as small as 100 pL [103]. Using a Photonic Professional GT printer (Nanoscribe GmbH, Germany) with a resolution finer than 1 μm^3 , they designed microfluidic channels integrated with a single-chip NMR probe, positioning samples such as tardigrade ova and nematode sections just 10 μm from the microcoil to optimize sensitivity and reduce static field inhomogeneities. These 3D-printed structures effectively guide and confine minute liquid and biological samples within the probe's most sensitive volume, delivering a sensitivity of 2.5×10^{13} spins/Hz $^{1/2}$ at 7 T and spectral resolutions as fine as 0.01 ppm, paving the way for non-invasive studies of single microscopic entities in biocompatible liquid environments. This rapid prototyping method surpasses the limitations of traditional microfabrication, providing a versatile and efficient platform for NMR investigations of sub-nL volumes essential to biological and medical research.

A novel mini-organic redox-flow battery sample chamber and probe head [104] were designed for in-situ and -operando NMR. This 3D-printed device was predominantly constructed via Polyjet 3D printing using polymethyl methacrylate (PMMA), a material with high transparency and magnetic susceptibility similar to water. The chamber was designed to plug onto a Bruker Micro-5 imaging probe, facilitating real-time monitoring of redox processes, by incorporating a microfluidic circuit driven by a mini bubble pump, ensuring continuous electrolyte flow within an 18 mm diameter form factor. The precise fabrication enabled by 3D printing allows for tailored components, like RF saddle coils and a detection area optimized for field homogeneity, supporting standard NMR experiments as well as magnetic resonance imaging and localized spectroscopy. This approach enabled monitoring of chemical reactions like dimerization during charge-discharge cycles, illustrating how 3D printing allows the creation of compact, customizable NMR tools to enhance the study of electrochemical systems with minimal sample volumes and high analytical precision.

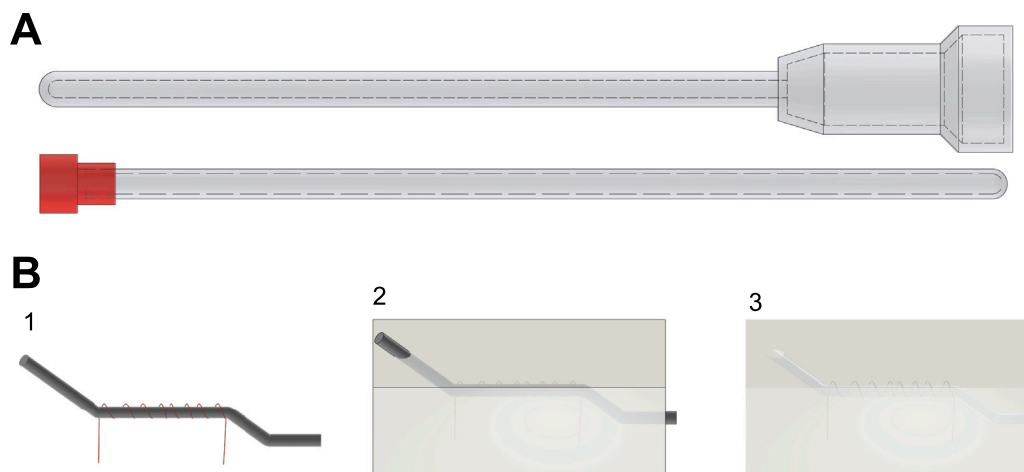


Fig. 6. (A) Gas-tight reaction vessels, including flasks, cuvettes, and NMR tubes fabricated via FDM 3D printing with ABS under inert N_2 conditions, taking advantage of its mechanical stability and low contamination risk, with ABS-K for reduced cuvette opacity [99]. (B) Shows ESCARGOT spiral microfluidic device that uses dissolvable ABS scaffolding in PDMS, cured and dissolved in acetone, enabling small-volume ^{31}P -coupled ^{19}F NMR analysis [101].

A sophisticated spin exchange optical pumping probe for hyperpolarizing xenon-129 gas was 3D printed using high-temperature plastics [105]. A Stratasys (Eden Prairie, MN) Fortus 360mc FDM printer, with a large build volume of $16 \times 14 \times 16$ in., was used to produce all of the components. The thermoplastic selected for this instrument was polycarbonate (PC) due to its tensile strength and heat deflection and glass transition temperatures, all of which are strong considerations for a device that has to operate at elevated temperatures and pressures. The 3D-printed structure seamlessly integrates critical components, including an in-situ NMR circuit, a narrowed 200 W laser, high-resolution near-IR spectroscopy, and thermoelectric temperature control, into a variable temperature probe, simplifying alignment and reducing production time and costs through rapid prototyping. This innovation achieves near-unity (almost 100%) ^{129}Xe polarization for a spherical phantom in a 0.5 L optical pumping cell, broadening the scope of hyperpolarization-enhanced gas imaging and NMR in chemical and biomedical research.

7. Sampling aid and packing tools

Sample insertion and retrieval in solid-state NMR experiments are often labor-intensive and time-consuming, especially when handling multiple samples daily. Particularly in the context of MAS DNP, with probes operating at temperatures in the tens of Kelvin, changing samples requires warming the probe, disconnecting cryogenic, sensor, pneumatic, RF, and microwave lines, removing the probe from the magnet, replacing the sample, and reversing the process – a sequence that could take hours, compounded by lengthy cooling cycles. Even rotor retrieval, though seemingly straightforward, poses challenges due to the tight tolerances and fragility of rotors and stators. Rotors typically sit flush within the stator housing, and improper handling risks damaging delicate components or misaligning the magic angle, necessitating recalibration. Although rotor eject mechanisms are now increasingly common, the first published 3D-printed component for NMR use appeared in 2009 as a rotor exchange mechanism for MAS DNP applications, streamlining sample changes and preserving instrument integrity [106]. Applicable to both cryogenic MAS DNP and room-temperature MAS experiments, it reduced mechanical wear and the risk of damaging fragile components like pin connectors during frequent probe handling. The design featured an N_2 gas-driven system with a custom 3D-printed angled eject pipe, fabricated using SLA printing with an ABS-like plastic, interfaced with a 4 mm Revolution NMR (Ft. Collins, CO) MAS stator. This pipe connected the probe to ejection tubing, guiding the rotor out the magnet's top without probe removal,

cutting both time and equipment stress. Advances in rotor ejection continued with Olson et al. [107], who 3D-printed a bent-tip extraction tool that mounts onto the probe body for seamless rotor removal. This suction-based system uses a bulb and flexible tubing to guide the rotor into a plastic sleeve at the tool's bend. Such innovations highlight how 3D printing enhances precision and adaptability in sample handling, addressing the mechanical challenges of MAS NMR.

Beyond rotor ejection, 3D printing has transformed rotor packing, a critical process that ensures dense, uniform sample filling to minimize air gaps, yielding consistent NMR signals and preserving sample integrity under high-speed spinning. Smaller rotors, such as commercial 0.7 mm outer diameter designs (holding sub- μ L sample volume due to wall thickness constraints), enable ultrafast MAS frequencies exceeding 100 kHz. These high speeds average out strong anisotropic interactions such as chemical shift anisotropy (CSA) and dipolar couplings that dominate NMR spectra of solids, while enhancing spectral resolution and SNR, particularly for complex biomolecules like proteins or nucleic acids. However, packing these diminutive rotors poses significant challenges, especially for biological solids. Samples ranging from fine, dry powders (e.g., crystalline peptides) to viscous, hydrated slurries (e.g., phospholipids or membrane proteins in liposomes) resist uniform filling due to their diverse physical properties, limited rotor volumes, and the risk of rotor wall stresses at extreme centrifugal forces, making precise tool design essential.

Ultracentrifugation has enabled packing of challenging samples into 3.2 mm and 4 mm rotors using custom PEEK-machined tools to endure high g-forces without rotor damage [108]. For 0.7 mm rotors, with a 0.5 mm inner diameter, SLA printing provides the precision for modular centrifugal packing devices [109]. Two 3D-printed tools, made with a Formlabs Form 3 and Tough 1500 resin, target biological solids: a low-relative centrifugal force (RCF) device for gentle sedimentation of sensitive samples and an ultracentrifugation device for dense packing. This tool uses interlocking connector pins and a keyway-locked base to compress an o-ring against the rotor collar, ensuring a seal while accommodating drive caps and spacers in open-ended rotors – though yielding looser packing due to reduced force. The ultracentrifugation tool, tested at 70,000 rpm (200,000 g) in a fixed-angle centrifuge, achieves tighter packing but risks leaks or rotor damage without proper sealing; it employs a rotor sleeve flush with a paraffin disk and O-ring, secured by snap-fitting tabs, omitting bottom spacers and drive tips to avoid cap compression (Fig. 7A). Post-printing, rotor holes were reamed with a 0.75 mm drill bit and flushed with isopropyl alcohol (IPA) for a precise fit. At 105 kHz MAS (11,000,000 g at the inner wall), ultracentrifuged 0.7 mm rotors further sediment dense

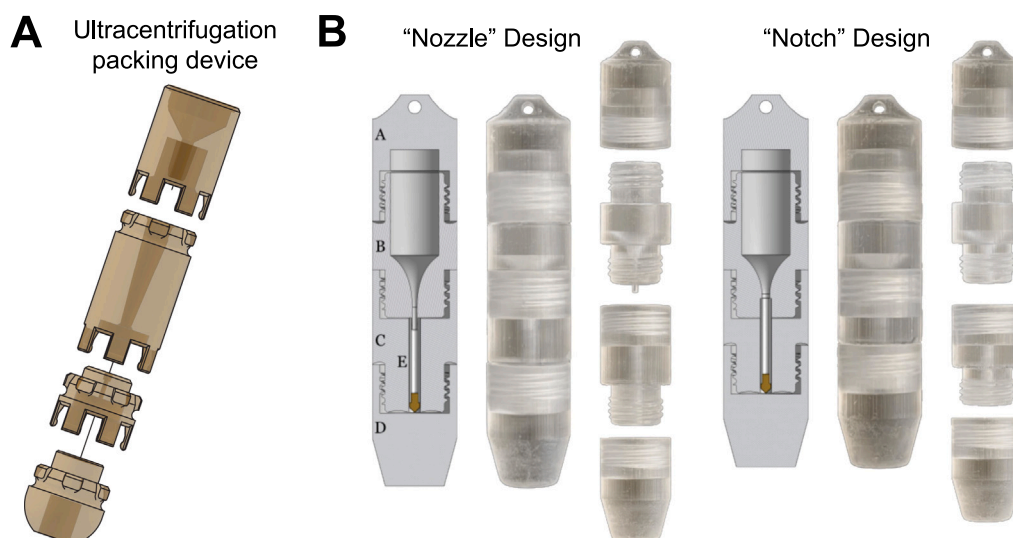


Fig. 7. (A) SLA 3D-printed packing tools made with Tough 1500 resin enable precise sedimentation of biological solids into 0.7 mm NMR rotors, with a low-RCF device for gentle packing and an ultracentrifugation tool achieving dense packing at 200,000 g, enhancing ^1H NMR signal ratios [109]. (B) Adapted from Hisao et al.'s Kel-F design, SLA-printed tools using Clear V4 and Durable V2 resins feature a hyperbolic 'nozzle' funnel or wider 'notch' for efficient sample transfer into 1.2–3.2 mm rotors, with modular components for unpacking and tachometer marking [107].

phases outward, leaving a supernatant or air column centrally. To mitigate this, rotors were packed via ultracentrifugation, spun at 105 kHz for an hour, then repacked similarly. Comparing ^1H NMR spectra of ^{15}N -labeled GB1 protein, $\text{NH}_2\text{H}_2\text{O}$ signal ratios were 0.3:1 (double-packed ultracentrifugation), 0.25:1 (single-packed ultracentrifugation), and 0.215:1 (low-RCF), demonstrating enhanced packing density with the 3D-printed ultracentrifugation tool. Modular low-RCF tools were also designed for Bruker (Billerica, MA) 1.3, 1.9, and 3.2 mm flat-bottom rotors and Phoenix NMR (Loveland, CO) 5 mm high-pressure rotors, tailored to specific centrifuges and Eppendorf tubes, with adaptable CAD files available [109]. All components distribute weight to minimize compressive stress, preventing rotor damage across packing methods.

Another packing tool design was inspired by and adapted from the one created by Hisao et al. [110], which was machined out of Kel-F, a chemically and thermally stable polymer that offers high rigidity. Similar to adapted devices discussed above, Hisao's packing tool CAD design, made for machining, cannot simply be used directly for 3D printing; modification was required. Olson et al. altered the original design to improve the tool's sample transfer capabilities and 3D printed with an SLA Form 3+ printer using either Clear V4 or Durable V2 resin, with z layer height resolution settings of 25 μm and 50 μm , respectively [107]. One of the improvements was to the funneling chamber that narrows the sample path into the rotor. It was changed from linear slope to a hyperbolic slope, which causes material to accelerate faster as it travels down to the very tip, reducing the chance of sample pelleting and clogging the tip. This funnel, called the "nozzle design", extends into the rotor and creates a seal for efficient sample transfer (Fig. 7B). For higher-viscosity samples, including hydrated fibrous protein aggregates, membrane proteins in liposomes, and microcrystalline samples, the "notch design" was implemented, with a much wider funnel tip (Fig. 7B). This design tightly grips the outside of the rotor and requires an additional O-ring to ensure a proper seal when transferring hydrated biomolecular samples. To showcase the versatility of this method and enable sample packing for a range of different instruments, they developed a comprehensive set of tools designed to support rotor sizes ranging from 1.2 mm to 3.2 mm, compatible with both Bruker and Varian/Phoenix NMR Pencil-style rotors. Moreover, due to threaded stackable components, they can be reconfigured to unpack the rotor via centrifugation and transfer the sample directly into another rotor. After a rotor is packed and ready for stator insertion, tachometer

markings are usually necessary. This entails manually marking one end of the rotor, usually with a Sharpie (Shelbyville, TN), where the fiber-optic cables measure spin rate. Here, the authors also 3D printed a tachometer marking guide to reproducibly mark 1.6 mm and 3.2 mm rotors.

Following the introduction of 3D-printed components to solid-state NMR, subsequent developments have expanded automation in sample handling for benchtop solution NMR. An autosampler reduces the need for human intervention, enhancing the efficiency introduced by eject systems. Manually changing NMR samples every few minutes demands continuous oversight, especially when performing NMR analysis on tens to hundreds of different samples. RotoMate – a fully functional 3D-printed autosampler – automates sample exchange, experiment initiation, and data interpretation for Magritek Spinsolve-series benchtop NMR spectrometers [111]. Integrated with Advanced Chemistry Development (ACD) NMR software, it holds up to 30 5 mm solution NMR tubes – outpacing the 20-tube capacity of typical commercial alternatives. A stepper motor, controlled by an Arduino microcontroller, drives its rotating "rotor" sample holder, while a servo motor regulates airflow in the pneumatic eject system. Safety features include a trapdoor to select samples precisely and a light sensor to confirm sample presence. This pneumatically controlled insert-and-eject system, encased in a custom 3D-printed housing, streamlines operation. The rotor, the largest component with disk-like geometry, risked warping from thermal contraction during FDM printing with PLA filament, potentially misaligning tubes radially. To address this, a PETG-printed motor flange and clutch ring were added to align the rotor horizontally, ensuring sample tubes align with the spectrometer's entry aperture; PETG's higher glass transition temperature preserved geometry near the warm motor. Other components, including the rotor and housing, were also FDM-printed with PLA. For precise sample positioning, a PLA "spinner" – mimicking high-field NMR spinners – was sleeved onto each tube's top, though FDM's tolerance limitations required post-processing to avoid stresses that could fracture glass tubes.

8. Visual aids

Many students, particularly beginning undergraduates, find visualizing objects in three dimensions challenging. Three-dimensional visualization aids such as molecular model kits and 3D-printed pdb structures have long been a beneficial pedagogical tool in chemistry to

translate two-dimensional drawings into the three-dimensional world. 3D printing has been used to construct visualization aids to better illustrate spectroscopic data and teach data analysis. Many spectroscopic techniques such as NMR and IR are instrumental in understanding a reaction's progression but can be difficult for students to analyze because there is a third component not easily represented on a two-dimensional plane of paper. Higman et al. used FDM printing to visualize reaction progress through printing data from the aforementioned spectroscopic techniques [112]. To create an STL file of an NMR spectrum, the data were exported as a CSV file, organized in a spreadsheet, and converted to an STL file in Mathematica. MeshLab and NetFabb were used for post processing of the STL file to fix any discrepancies and resize. An isomerization reaction was monitored by 1D ^1H NMR and this was visualized by printing a display correlating ppm, reaction time, and intensity as the x , y and z coordinates [112]. Similarly, a separate method was developed to convert 1D and 2D NMR data into STL files by exporting as a text file, processing with Python, and converting to STL with Mathematica. A TOCSY spectrum of thymidine and inversion-recovery T_1 measurement for 1-hexanol were printed using FDM to show the peaks' positions and intensities [113]. For individual data sets, optimization may be required if the peak intensities are too sharp, decreasing their structural stability. Lorentzian and Gaussian functions were used to artificially thicken the models, with Lorentzian being more accurate but less sturdy than Gaussian [113]. A third group developed methods for 3D printing NMR spectra and HPLC chromatograms for both FDM and SLA printing. For NMR spectral processing, they exported the data as a text file, processed with Python, and generated the STL file with MATLAB. A graphical user interface in MATLAB was coded to process and scale the Python data to the desired size, then convert into a STL file. An HSQC and a COSY spectrum were printed using PLA and UV curable resin for FDM and SLA printing respectively [114].

Although these approaches are mostly limited in practice to data sets that are all-positive or at least continuously varying like the inversion-recovery, they are useful for helping students begin to visualize the spectra in 3D. These visualization aids can help illustrate the topology of an NMR spectrum and how to interpret it. Students can see the decrease/increase in peak intensity over a reaction course [112] and connect a specific functional group's role in the reaction. 3D printed NMR spectra [113,114] can be used in conjunction with 3D printed protein structures [115] to correlate peaks to specific residues on the protein. In all papers mentioned, supplementary information is provided with detailed instructions for data processing along with any scripts/files needed to convert data using the method described, making these tools implementable in teaching lectures and labs that have access to a 3D printer.

9. Conclusion

The expanding use of 3D printing in NMR instrumentation empowers both experienced and novice designers by enabling low-cost, tailored components with intricate geometries. For established groups, these technologies reduce barriers to prototyping innovative designs, encouraging collaboration through shared CAD files and improving reproducibility via standardized fabrication across labs. For researchers without access to traditional machining skills or infrastructure, 3D printing offers an accessible pathway to create bespoke NMR components – whether in durable plastics, high-conductivity copper, or beyond – unlocking specialized applications without significant capital investment, thus democratizing instrument development and accelerating field-wide progress. As 3D printing evolves, future efforts will extend beyond mere durability and spatial resolution, targeting material properties like electrical conductivity, thermal stability, and dielectric insulation for radiofrequency applications critical to NMR probes.

CRedit authorship contribution statement

Jose L. Uribe: Writing – review & editing, Writing – original draft, Visualization, Investigation, Conceptualization. **Annie V. McAllister:** Writing – review & editing, Writing – original draft, Visualization, Investigation, Conceptualization. **Rachel W. Martin:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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Glossary of abbreviations

3D: three-dimensional
ABS: acrylonitrile butadiene styrene
ACD: Advanced Chemistry Development
AM: additive manufacturing
CAD: computer-aided design
CFS: continuous flow separation
CNC: computer numerical control
CP: cross-polarization
CSA: chemical shift anisotropy
CSV: comma-separated value
DIY: do-it-yourself
DLP: digital light processing
DNP: dynamic nuclear polarization
ESCARGOT: Embedded SCAfold RemovinG Open Technology
FDM: fused deposition modeling
FFF: fused filament fabrication
IACS: International Annealed Copper Standard
IR: infrared
LCD: liquid crystal display
MaDaMEPro: Material Database for Magnetic Properties
MAS: magic angle spinning
MRI: magnetic resonance imaging
MSLA: masked stereolithography

NMR: Nuclear Magnetic Resonance
PA: polyamide
PBF: powder bed fusion
PC: polycarbonate
PCTFE: polychlorotrifluoroethylene
PDMS: polydimethyl-siloxane
PEEK: polyether ether ketone
PETG: polyethylene terephthalate glycol
PHIP: para-hydrogen induced polarization
PLA: polylactic acid
PMMA: poly-methyl methacrylate
PPE: personal protective equipment
PVDF: polyvinylidene fluoride
PTFE: polytetrafluoroethylene
RCF: relative centrifugal force
RF: radiofrequency
SABRE: Signal Amplification by Reversible Exchange
SDS: safety data sheet
SLA: stereolithography
SLM: selective laser melting
SLS: selective laser sintering
SNR: signal-to-noise ratio
STL: standard tessellation language
TOCSY: T₀tal Correlation Spectroscopy
UV: ultraviolet
VOC: volatile organic compound